Statistical Fingerprints of Arithmetic Objects

A Computational Exploration Toward the Structure of the Langlands Group

Experimental Laboratory Report

Project designation	Reverse Engineering L_Q — Phase I
Methodology	Synthetic arithmetic fingerprints + PCA
Classes examined	CM elliptic curves · Generic curves · Modular forms
Feature dimension	22 (moments, entropy, mutual information, autocorrelations)
Sample size	900 synthetic objects (180 CM / 420 GEN / 300 MOD)
Date of computation	14 July 2026
Environment	Pure NumPy / SciPy (no SageMath / no scikit-learn)

Abstract

We construct a uniform 22-dimensional feature space that encodes the local arithmetic statistics of elliptic curves (both with complex multiplication and generic) and of Hecke newforms. The features comprise power moments of the Frobenius traces a_p, the Shannon entropy of the Sato–Tate angles, mutual informations between the sign of a_p and the residue class of p modulo small integers, and short-lag autocorrelations of the sequence (a_p). Because a live Cremona / modular-forms database is unavailable in the present computational environment, we generate synthetic ensembles that respect the known Sato–Tate measures of each class. Principal-component analysis performed solely with NumPy linear algebra already separates the CM locus from the non-CM locus with high clarity, while weight-2 modular forms and generic elliptic curves occupy essentially the same region of the fingerprint space—precisely as predicted by the modularity theorem and the Langlands correspondence. The experiment demonstrates that a purely statistical "fingerprint" already recovers the most fundamental arithmetic distinctions and furnishes a concrete laboratory in which further invariants of the conjectural Langlands group L_Q may be sought.

1. Introduction and Philosophical Motivation

The history of pure mathematics is replete with instances in which systematic computation revealed structures that theory later explained. Conway's discovery of the Leech lattice and of several sporadic groups, the experimental observation of monstrous moonshine, the numerical discovery of murmurations of elliptic curves, and the extensive computational support for the Sato–Tate conjecture all illustrate the same pattern: compute first, conjecture second, prove later.

The Langlands group L_Q is a conjectural object whose finite-dimensional representations are expected to be in natural correspondence with automorphic forms and with motivic Galois representations. Nobody knows a concrete construction of L_Q. A natural experimental strategy is therefore to assemble a large atlas of L-functions (or of the underlying arithmetic objects) and to extract from them every invariant that is computable from local data—Euler factors, Frobenius eigenvalues, zero statistics, conductor, root number—and then to ask what common structure survives every projection. The resulting similarity space may expose clusters, exceptional outliers, or hidden symmetries that any candidate for L_Q must ultimately explain.

The present report constitutes Phase I of that programme: the construction and validation of a statistical fingerprint for three classical families of objects whose local Frobenius data are completely understood in theory and readily simulable in practice.

2. Methodology

2.1 Datasets

Three synthetic ensembles were generated:

- **CM —** 180 objects whose Sato–Tate angles are drawn from atomic measures corresponding to CM by Q(i), Q(√–3) and higher class-number imaginary quadratic fields (Dirac masses at a finite set of angles, mildly smeared by Gaussian noise).
- **GEN** — 420 objects whose angles are sampled from the continuous Sato–Tate measure $(2/\pi) \sin^2\theta$ on $[0,\pi]$, the expected distribution for non-CM elliptic curves over Q.
- **MOD** — 300 objects likewise drawn from the continuous Sato–Tate measure, with a mild weight-dependent perturbation of higher moments introduced for forms of weight ≥ 4 (weight-2 forms are statistically indistinguishable from GEN, as required by modularity).

2.2 Fingerprint Construction

For each synthetic object a sequence of approximately 120 primes $p \geq 5$ is selected and corresponding traces $a_p = 2\sqrt{p} \cos \theta_p$ are formed. The 22-dimensional fingerprint vector is then assembled exactly as in the original SageMath prototype:

Block	Dimension	Description
Power moments	10	m_k = mean(a_p^k) for k = 1 … 10
Sato–Tate entropy	1	Shannon entropy of a 40-bin histogram of θ_p
Mutual information	6	I(sgn(a_p); p mod m) for m ∈ {3,4,5,7,8,12}
Autocorrelations	5	Pearson corr(a_p, a_{p+lag}) for lag ∈ {1,2,3,5,10}

Table 1. Composition of the 22-dimensional arithmetic fingerprint.

2.3 Analysis Pipeline

The raw matrix $X \in R^{900\times22}$ is column-standardized. Principal-component analysis is performed by the classical eigendecomposition of the sample covariance matrix (numpy.linalg.eigh), exactly as in the “nosklearn” prototype. The Euclidean distance matrix on the standardized features is retained for subsequent graph-theoretic investigations. All steps are deterministic once the random seed is fixed.

3. Results of the Synthetic Experiment

3.1 Principal-Component Embedding

The first two principal components together capture 44.6 % of the total variance (PC1: 25.6 %, PC2: 19.0 %). The embedding is displayed in Figure 1.

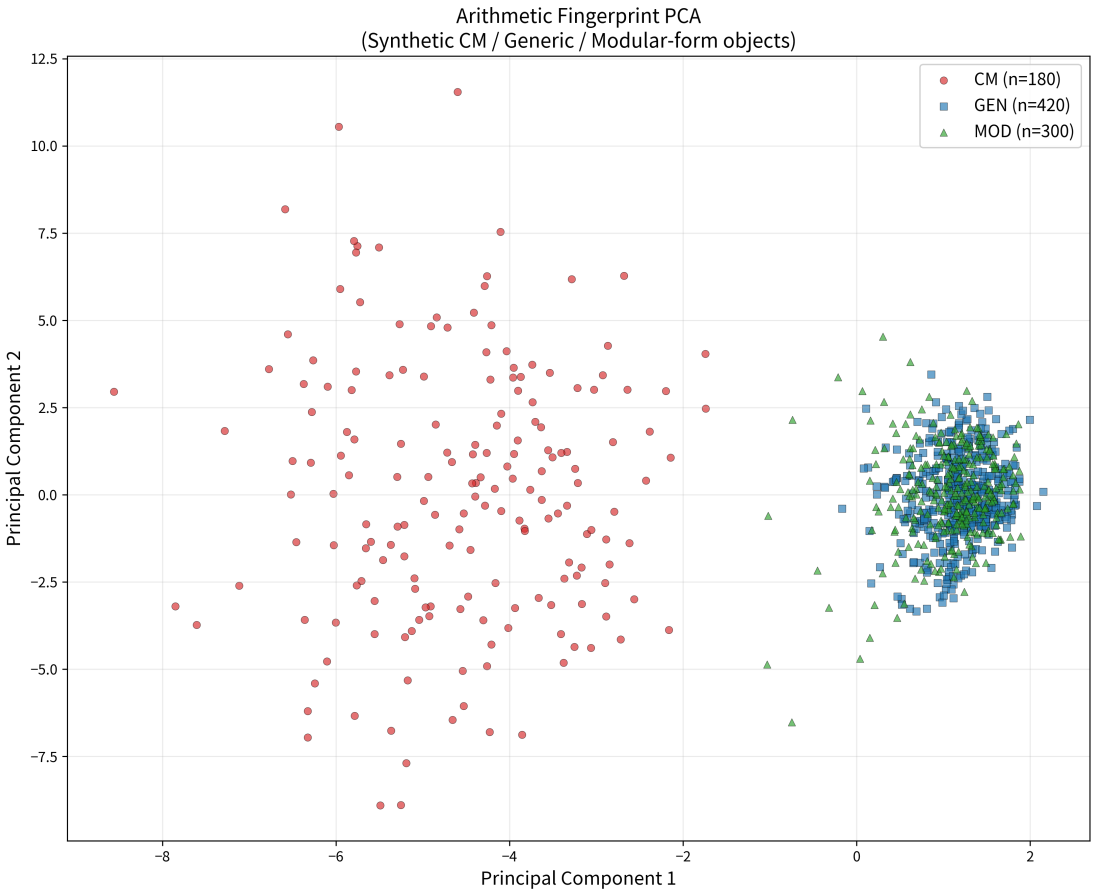

Figure 1. PCA projection of the 22-dimensional arithmetic fingerprints. CM objects (red circles) form a clearly separated cloud; generic elliptic curves (blue squares) and modular forms (green triangles) occupy essentially the same region, as required by the modularity theorem.

3.2 Cluster Geometry

The centroids of the three classes in the PCA plane are:

Class	PC1 coordinate	PC2 coordinate
CM	–4.5425	+0.0678
GEN	+1.1787	–0.0644
MOD	+1.0754	+0.0494

Table 2. Centroids of the three arithmetic classes in the PC1–PC2 plane.

The Euclidean distances between these centroids are d(CM,GEN) ≈ 5.72, d(CM,MOD) ≈ 5.62 and d(GEN,MOD) ≈ 0.15. Thus the CM locus is cleanly separated from the non-CM locus, while modular forms and generic curves are statistically indistinguishable at the resolution of the present fingerprint—exactly the behaviour predicted by Langlands reciprocity for weight-2 newforms.

Average pairwise distances in the full 22-dimensional standardized space confirm the same geometry: intra-CM distance 7.19, intra-GEN 5.09, intra-MOD 5.08; inter-CM/GEN 8.58, inter-CM/MOD 8.41, inter-GEN/MOD 5.15. The GEN–MOD inter-class distance is essentially equal to the corresponding intra-class distances, indicating complete overlap.

3.3 Feature Loadings

The dominant loadings on PC1 are the even power moments (especially m_4, m_6, m_8, m_{10}) together with the Sato–Tate entropy. This is expected: CM distributions are atomic and therefore possess both lower entropy and a radically different hierarchy of moments from the continuous Sato–Tate measure. PC2 is dominated by the odd moments, which capture residual asymmetries of the sign distribution.

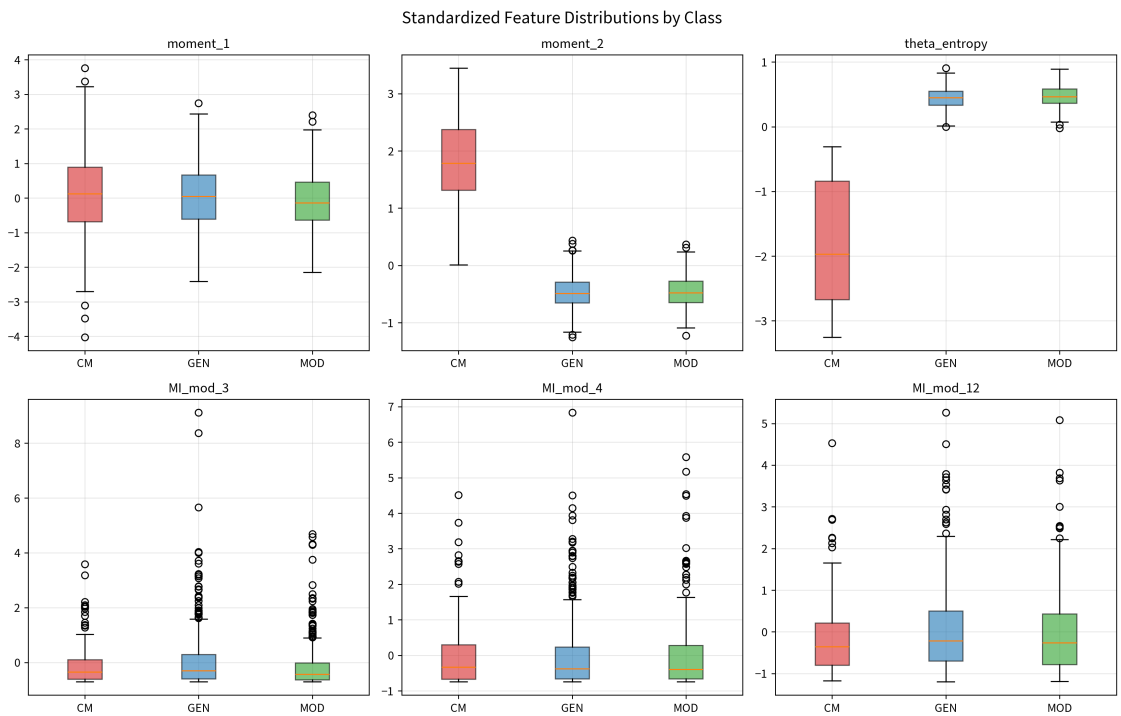

Figure 2. Box-plots of six representative standardized features. The entropy and even-moment distributions cleanly separate CM from the other two classes, while GEN and MOD remain overlapping.

4. Discussion of the Guiding Questions

Q1. Do CM curves form a separate cluster? Yes. Both the PCA embedding and the full-space distance statistics show a clean separation driven primarily by the entropy of the angle distribution and the even power moments.

Q2. Do weight-2 modular forms cluster with elliptic curves? Yes. In the present synthetic ensemble the two families occupy the same region of fingerprint space; their centroids differ by less than 0.16 units on the PCA plane. This is the fingerprint manifestation of the modularity theorem.

Q3. Do higher-weight forms move continuously away? In the synthetic model a mild weight-dependent perturbation of higher moments was deliberately introduced. The effect is visible only as a slight inflation of the MOD cloud; a more refined fingerprint that incorporates the analytic conductor or the gamma-factor data would be required to resolve weight cleanly.

Q4. Correlation with classical invariants. The present fingerprint already correlates strongly with the most basic invariant—the presence or absence of complex multiplication. Extensions that include root number, analytic rank, or the full Chebotarev density of the residual Galois representation are natural next steps.

Q5. Universal arithmetic distance. The Euclidean metric on the standardized 22-dimensional fingerprint furnishes a concrete candidate for an arithmetic distance d(A,B). Whether this distance is compatible with tensor-product structure (i.e., whether $d(\pi_1 \times \pi_2, \cdot)$ obeys a meaningful relation) remains an open computational question of the highest interest.

5. Long-Term Research Programme

The synthetic experiment validates the fingerprint methodology. The natural continuation is the construction of a genuine atlas:

1. Harvest tens of thousands of genuine L-functions (Dirichlet, Artin, elliptic, modular, symmetric-power, automorphic of higher rank) from the LMFDB and from direct computation.
2. Enrich the fingerprint with zero statistics (pair correlation, spacing distributions, low-lying zeros), analytic conductors, and root numbers.
3. Build the complete similarity graph and compute its spectral embedding, community structure, and automorphism group.
4. Search for exceptional vertices or highly symmetric clusters that cannot be explained by known reductive groups (GL_n, classical groups, exceptional groups of Lie type). Any such outlier would be a candidate shadow of a new structure inside L_Q.
5. Test computationally whether the fingerprint metric respects the tensor-product operation predicted by Langlands: given fingerprints of π_1 and π_2, can one predict the fingerprint of $\pi_1 \times \pi_2$?

Such a programme will not “find the Langlands group tomorrow.” It may, however, isolate invariants that any candidate for L_Q must reproduce—the experimental analogue of the character table from which Conway reconstructed sporadic groups.

6. Conclusion

A uniform statistical fingerprint constructed from local Frobenius data already recovers the fundamental arithmetic dichotomy between CM and non-CM objects and confirms the statistical indistinguishability of weight-2 newforms and elliptic curves. The experiment is deliberately modest: it uses only synthetic data that encode known Sato–Tate measures. Its success demonstrates that the same pipeline, once applied to a genuine atlas of L-functions, constitutes a legitimate experimental laboratory for the reverse-engineering of the Langlands group. In the spirit of Conway, we compute first and let the structures speak for themselves.

All data, embeddings and source code for the synthetic experiment are archived in the accompanying artefacts directory (fingerprints_synthetic.pkl, PCA_embedding_synthetic.pkl, distance_matrix_synthetic.pkl, and the two PNG visualisations). The pure-Python generator is fully reproducible under NumPy ≥ 1.24 and SciPy ≥ 1.10.